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ON LOCALIZATION IN THE THEORY OF THE QUANTIZED HALL EFFECT: A TWO-DIMENSIONAL REALIZATION OF THE θ-VACUUM

A M M PRUISKEN

Schlumberger-Doll Research, PO Box 307, Ridgefield, CT 06877, USA

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It is shown that the localization problem in the theory of the quantized Hall effect is governed by the zero-component grassmannian U($2m$) non-linear σ-model with a θ-term, a two-dimensional analogue of the θ-vacuum in Yang-Mills theory In this case, θ is to be interpreted as the "bare" value for the Hall conductivity, determined by an underlying non-critical theory A detailed derivation is presented starting from the replica method and a delta function distribution for the impurities

1. Introduction

In this paper I apply and extend the techniques developed in refs. [1, 2] for the analysis of the Anderson model, in order to discuss the disordered electronic system subject to a magnetic field. Especially in two dimensions the problem is of considerable interest. In the case of relatively strong magnetic fields (applied perpendicular to the system) and relatively weak disorder, experiments show a quantized behaviour of the Hall conductance as a function of electron density (quantized Hall effect). The usual theoretical explanation of this phenomena refers to the effect of electron localization: in a certain region between successive Landau levels only localized electron states exist which cannot contribute to the conduction, outside this region extended states must exist in order for the quantized Hall effect to occur.

As such, the problem is challenging the "established" theory on localization which predicts a mobility edge in $2 + \varepsilon$ dimensions whereas in $d = 2$ only localized states should occur. Indeed, there have been suggestions made [11, 12] that the system under consideration is described by the zero component U($2m$) non-linear σ-model for which these conclusions certainly apply. However it has been argued on general grounds that these conclusions of localization theory most likely do not apply to the system under consideration [8]; based on concepts of gauge-invariance and energy conservation in the process of electron transport, an argument has been constructed

leading convincingly to the existence of extended states in the present system for sufficiently small impurity concentration [7,6].

In what follows, I reconsider the non-interacting problem with a white-noise distribution for the random potential. As a starting point of the discussion I use an effective medium approximation for the density of states which has been studied previously by Ando [16]. As in the original Anderson model I find that, within perturbation theory, the density of states varies continuously with energy and does not enter the theory as a scaling parameter (see also [13]). Long-ranged behaviour in the averaged one-particle Green function does exist however if one imposes boundaries on the system. This aspect is closely related to Halperin's observation [7] of current carrying states near the sample boundary which persist also in the presence of disorder. In the context of ensemble averaged Green functions this insight is gained from an interpretation of the current as an operator acting on the sample boundaries.

In formulating an effective lagrangian for the disordered system, I find a new term which appears with a strength equal to the "bare" or unrenormalized Hall conductance. This term is the topological invariant of the $U(m+n)/U(m)\times U(n)$ non-linear σ-model [18,19] which essentially can be expressed as a total derivative and which therefore cannot be detected by the perturbative renormalization group analyses underlying all work on electron localization thus far.

The physical meaning of this new term is contained in the following two results which will be published elsewhere [9]. First it can be shown as a direct consequence that the Hall conductance is quantized in units e^2/h for an exponentially localized state. Second, it can be shown that this term is responsible for the occurrence of extended states in direct analogy with the physics of the θ-vacuum in 4d Yang-Mills theory.

This paper is organized as follows.

Sect. 2 briefly describes the formalism, i.e. the replica method for performing the averaging over a gaussian distributed random potential and underlines the basic assumptions involved.

Sect. 3 deals in detail with the derivation of the effective lagrangian in terms of compact and non-compact symmetries. The breaking of time-reversal invariance by the magnetic field manifests itself (on a microscopic level) in the crossing over from orthogonal (zero field) to unitary (non-zero field) symmetry between the replicated advanced and retarded waves. The general methodology is then very much the one outlined in [1,2]: namely the introduction of matrix or composite fields, the formulation of a mean field theory which incorporates the unitary symmetry in the problem and which gives rise to a natural separation between the massive field-components and the Goldstone modes. One then carefully proceeds by eliminating the massive fields while retaining the massless ones. However, the constant magnetic field has the additional aspect of breaking parity (or rather: pseudo parity, namely the symmetry $x \to -x$ and $y \to -y$ separately) underlying the Onsager relation for the Hall conductance. This aspect reflects the time-reversal non-invariance on a

macroscopic level and manifests itself most clearly in the effect of imposing boundaries on the system: namely the existence of current carrying states (long-ranged behaviour in the averaged one-particle Green function) near the sample-boundary and the appearance of a (pseudo) parity breaking boundary term in the effective lagrangian.

Finally (sect. 4) a discussion is given on the results so far.

2. Starting point

We wish to study the effect of disorder on a two-dimensional, non-interacting electron gas in a magnetic field B in the z-direction, perpendicular to the system. The hamiltonian is

$$H = \tfrac{1}{2}m(\nabla/\imath + e/cA)^2 + V(r) = \pi_\mu \pi^\mu + V(r), \tag{2.1}$$

where $V(r)$ is a random potential which for simplicity is distributed according to

$$P[V(r)] = \mathfrak{N}^{-1}\exp\left\{-\frac{1}{2g}\sum_r V^2(r)\right\}. \tag{2.2}$$

To represent the magnetic field we choose the Landau gauge

$$A = (0, Bx, 0), \tag{2.3}$$

$$B = (0, 0, B). \tag{2.4}$$

2.1 GENERATING FUNCTION

Instead of looking at one particular disordered sample one considers an ensemble average for the Green functions $(E - H)^{-1}$. For these quantities a generating function can be constructed upon employing the replica method [1–3]. In order to do so one notices that H is a hermitian operator rather than a real-symmetric one (which is true for $B = 0$). This determines the functional integrals to be over complex fields rather than real ones. Introducing a complex energy

$$E_p = E + \imath s_p \eta, \qquad s_p = (1, -1), \tag{2.5}$$

where η is positive infinitesimal and where the index p distinguishes between the advanced and retarded Green functions we consider a partition function

$$Z = \int \mathrm{D}[V]\, P[V] \int \mathrm{D}[\bar{\Psi}]\, \mathrm{D}[\Psi] \exp L(\bar{\Psi}, \Psi, V),$$

$$L = \sum_r \sum_{p=1}^{2} \sum_{a=1}^{m} \bar{\Psi}_a^p(r)(E_p - H)\Psi_a^p(r). \tag{2.6}$$

The index a runs over m replicas where in the end $m \to 0$. For example the averaged one-particle Green function is given as

$$G_p(r,r') = \overline{\langle r|(E_p - H)^{-1}|r'\rangle}$$

$$= \pm\langle \bar{\Psi}_a^p(r)\Psi_a^p(r')\rangle \begin{cases} +\text{fermion representation} \\ -\text{boson representation.} \end{cases} \qquad (2.7)$$

The brackets $\langle\ \rangle$ indicate the statistical average with respect to L. At this stage, one is still free to specify the Ψ-fields to be of the boson or fermion (Grassmann) type. In the former case the $\bar{\Psi}^p$ is defined as $is_p\Psi^{p+}$ such that $\eta > 0$ in E_p assures the convergence of the integrals. This then gives rise to non-compact symmetries: in the limit $\eta \to 0$, L is invariant under transformations which leave the form $\Psi_a^{+(1)}\Psi_a^{(1)} - \Psi_a^{+(2)}\Psi_a^{(2)}$ invariant, i.e. under transformations contained in the non-compact group $U(m,m)$. Subsequent analysis will be mainly done in boson representation, in order to make contact with refs. [1,2,4]. However, it could equally well be performed in terms of complex Grassmann fields, in which case $\bar{\Psi}, \Psi$ anticommute and $\bar{\Psi} = i\Psi^+$. Then the underlying symmetry of L is the compact group $U(2m)$, broken by $\eta \neq 0$ down to the level $U(m) \times U(m)$.

The two representations for the generating function are genuinely different for non-zero m. The answers one gets in the analytic continuation $m \to 0$ should be the same however. We will return to this question at a later stage.

2.2 LOCALIZED VERSUS EXTENDED STATES

As a general remark, the importance of the continuous symmetries for the localization problem stems from Ward identities which can be derived from them and which maintain their validity in the limit $m \to 0$. In particular

$$\sum_{r'} \overline{G_1(r,r')G_2(r',r)} = \frac{i}{2\eta}\left[\overline{G_1(r,r)} - \overline{G_2(r,r)}\right] = \frac{\pi}{\eta}\rho(E) \qquad (2.8)$$

can be obtained from an infinitesimal $U(m,m)$ ($U(2m)$) rotation on Ψ [3]. The density of states $\rho(E)$ plays a special role in this context: on the one hand it can be understood as a local variation in the symmetry breaking term η

$$\rho(E) = \frac{i}{2\pi}\left[\overline{G_1(r,r)} - \overline{G_2(r,r)}\right] = \frac{-1}{(2\pi m)}\partial \ln(Z)/\partial\eta(r), \qquad (2.9)$$

and hence the notion of the density of states as an order parameter.

On the other hand, that is in conventional problems for electronic disorder, $\rho(E)$ does not show any criticality at all. I will assume for the present system $\rho(E)$ also to be a harmless, continuous parameter, or more generally there is no critical behaviour

in Green functions involving one of the two subspaces $p = 1, 2$ only. Justifications of this claim will be given in subsect. 3.2.

Eq. (2.8) then contains the localization theorem as an alternative to the Goldstone theorem [14]: for an extended state the divergence on the right-hand side of eq. (2.8) originates from the long-ranged part of the two-particle Green function; for a localized state the divergence comes from the terms with $r \approx r'$ which in turn can be identified with the inverse of the volume fraction taken by that state [3].

2.3 THE EFFECT OF IMPOSING BOUNDARIES

There is however an important difference between the one-particle Green function theory in the Anderson problem and the disordered electronic system subject to a magnetic field. Namely the existence of long-ranged behaviour in the averaged one-particle Green function near the sample boundary. This observation has direct consequences for the field theory, which is still to be discussed, and ultimately for the behaviour of a new physical quantity in the problem, namely the Hall conductance or σ_{xy}.

In ref. [7] it has been shown that in the system without disorder and in case the Fermi energy is located in a gap, the appearance of the (quantized) Hall conductance can entirely be understood from the existence of a current carrying state (at the Fermi energy) near the boundary of the sample. It has been argued [7] that this edge effect also persists for the disordered system. In subsect. 3, I essentially confirm this statement by studying (as in [7]) the expectation of the current at the Fermi level

$$J_\mu(E, r) = \sum_i \Psi_i^+(r)\left[\pi_\mu + \pi^\mu\right]\Psi_i(r)\delta(E - e_i), \tag{2.10}$$

where Ψ_i is the wave function of the disordered sample with energy e_i. As a direct translation in an ensemble average this quantity becomes

$$\overline{J_\mu(E, r)} = \frac{1}{2\pi i}\overline{\left[\pi_\mu + \pi^\mu\right]\left[G_1(r, r) - G_2(r, r)\right]}\,, \tag{2.11}$$

which is a typical one-particle Green function quantity. In sect. 3 it will be shown that this quantity acts like an operator in the following sense

$$\int \overline{J_\mu(r)}\,\Omega(r)\,\mathrm{d}^2 r = \int\left[\sigma_{xy}^{\mathrm{II}}\varepsilon_{\mu\nu}\nabla_\nu + \mathrm{O}(\nabla^2)\right]\Omega(r)\,\mathrm{d}^2 r, \tag{2.12}$$

for an arbitrary smooth function $\Omega(r)$.

$$\sigma_{xy}^{\mathrm{II}} = \int_{-\infty}^{E} \mathrm{d}E\,\frac{\partial}{\partial B}\rho(E). \tag{2.13}$$

The quantity σ_{xy}^{II} is the exact "non-classical" contribution to the Hall conductance in the sense of Streda [5] and is a property of the interior of the system.

The operator statement above results from a systematic expansion in powers of the gradient and holds over length scales larger than the phase-coherence length, which determines the range of the one-particle Green function in the interior of the system. Formally, we can write

$$\overline{J_\mu(r)} = \sigma_{xy}^{II}\varepsilon_{\mu\nu}\nabla_\nu + O(\nabla^2). \tag{2.14}$$

The meaning of this equality can be studied via a simple example. Namely, the effect of imposing boundaries on the system can be incorporated specifying the function $\Omega(r)$ to be unity in the interior of the sample and zero elsewhere; the cross-over between these values is thought of as being smoothed out over a length, approximately equal to the phase-coherence length. Then clearly, the integrant above is non-zero only near the edge of the sample. The equation above tells us that a current is flowing along the perimeter of the system and the amount (per energy) through the surface perpendicular to an edge equals σ_{xy}^{II}. This then justifies the statement made above, namely the presence of long-ranged behaviour in the one-particle Green function near the sample boundary.

Notice furthermore, that the equation above relates a bulk property (namely the expression for σ_{xy}^{II}) to the occurrence of a boundary effect. According to linear response theory, the Hall conductance can be expressed as the sum of σ_{xy}^{II} and another ("classical") contribution σ_{xy}^{I} [5]. In ref. [10] it will be shown that the extra term σ_{xy}^{I} is entirely due to the interaction of the system with the electric field.

2.4 THE CROSSOVER PROBLEM

Next, in order to make contact with the Anderson problem, we split the field Ψ into its real and imaginary components

$$\Psi_a^p = \sqrt{\tfrac{1}{2}}\,(\Theta_a^p + i \times \Phi_a^p), \tag{2.15}$$

then L (2.6) can be written

$$L = \tfrac{1}{2}\sum \bar{\psi}_a^p(r)(E_p - \hat{H})\psi_a^p(r), \tag{2.16}$$

where ψ indicates the spinor

$$\psi_a^p = \begin{bmatrix} \Theta_a^p \\ \Phi_a^p \end{bmatrix} \begin{cases} \bar{\psi} = is_p\psi^T, & \text{boson representation} \\ \bar{\psi} = -\psi^T\tau_2, & \text{fermion representation}, \end{cases} \tag{2.17}$$

and $\hat{H}$ is given by

$$\hat{H} = \frac{-1}{2m}\left(\nabla^2 - \frac{e^2}{c^2}A^2\right)\tau_0 + V(r)\tau_0 + i\frac{e}{mc}A_\nu\nabla_\nu\tau_2, \tag{2.18}$$

with τ_i the Pauli matrices

$$\tau_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \tau_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\tau_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \qquad \tau_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{2.19}$$

Clearly, when $A = 0$ we recover the formalism used for discussing the Anderson problem. The introduction of the spinor ψ in the boson representation merely doubles the number of replica channels from m to $2m$ and the underlying symmetry is $O(2m, 2m)$. Non-zero A breaks the symmetry down to $U(m, m)$. In the fermion representation the corresponding symmetry is the symplectic group $Sp(4m)$ which by a magnetic field is broken down to $U(2m)$. This is a direct consequence of the breaking of time-reversal invariance by the magnetic field.

3. Effective field theory

3.1 COMPOSITE FIELDS

The average over the gaussian distribution $P[V]$ in (2.6) can be carried out explicitly, leading to a lagrangian in terms of $\bar{\Psi}, \Psi$ only

$$L \to \sum \bar{\Psi}_a^p(r)\left[E_p - \pi_\mu \pi^\mu\right]\Psi_a^p(r) + \tfrac{1}{2}g\sum_r\left[\sum_{p,a}\bar{\Psi}_a^p(r)\Psi_a^p(r)\right]^2,$$

$$\bar{\Psi}_a^p = is_p\Psi_a^{*p}. \tag{3.1}$$

The four-point term in (3.1) can be decoupled via the introduction of a matrix field $Q_{ab}^{pp'}$ replacing the combination $\bar{\Psi}_a^p\Psi_b^{p'}$. The equivalent of eq. (2.6) then reads

$$Z = \int \mathrm{D}[Q]\int \mathrm{D}[\bar{\Psi}]\,\mathrm{D}[\Psi]\exp\left[L(\bar{\Psi}, \Psi, Q)\right], \tag{3.2}$$

$$L = -\frac{1}{2g}\sum Q^2 - \frac{i}{g}\sum \eta s Q + \sum \bar{\Psi}_a^p(r)\left[\left(E - \pi_\mu\pi^\mu\right)\delta_{ab}^{pp'} - Q_{ab}^{pp'}(r)\right]\Psi_b^{p'}(r). \tag{3.3}$$

In order to preserve the symmetries it is required that Q is invariant under $Q \to T^{-1}QT$ with $T \in U(m, m)$, i.e. pseudo-hermitian

$$Q = sQ^+ s. \tag{3.4}$$

The various aspects of this transformation (in particular convergence) have been discussed in detail in [4, 2], here we simply state that the validity of the Ψ as well as the Q integrals is entirely due to $\eta > 0$. The expectation of the field Q has a simple physical meaning, namely

$$\langle \bar{\Psi}_a^p(r)\Psi_b^{p'}(r)\rangle = \frac{-1}{g}\langle Q_{aa}^{pp}(r)\rangle = \big(e(E) + \imath\pi S_p \rho(E)\big)\delta_{ab}^{pp'}. \tag{3.5}$$

The expectation of the matrix Q is given in terms of the exact density of states $\rho(E)$ (as $m \to 0$). One now proceeds by integrating out the $\bar{\Psi}, \Psi$ in (4.3) in order to obtain a formulation in terms of the Q only

$$L \to -\frac{1}{2g}\operatorname{Tr} Q^2 - \frac{\imath}{g}\operatorname{Tr}\eta s Q - \operatorname{Tr}\ln\big[E - \pi_\mu\pi^\mu - Q_{ab}^{pp'}(r)\big]. \tag{3.6}$$

In the analysis of (3.6) we start out employing the saddle-point method. Minimizing L with respect to the fields Q we find the saddle-point equation

$$Q_{sp} = q_a^p \delta_{ab}\delta_{pp'},$$

$$\frac{1}{g}q_a^p + \frac{i}{g}\eta s_p - \langle r|\big(E_p - \pi_\mu\pi^\mu - q_a^p\big)^{-1}|r\rangle = 0. \tag{3.7}$$

The matrix inverse in (3.7) can be evaluated if we make use of the eigenfunctions and values of the operator $\pi_\mu\pi^\mu$

$$\pi_\mu\pi^\mu\xi_{n,k}(r) = \omega_c\big(n + \tfrac{1}{2}\big)\xi_{n,k}(r) = \varepsilon_n\xi_{n,k}(r), \tag{3.8}$$

where

$$\xi_{n,k}(r) = \mathrm{e}^{\imath k y}\phi_n(x + k\alpha). \tag{3.9}$$

ϕ_n are the harmonic oscillator functions, $\omega_c = eB/mc$ the cyclotron frequency and $\alpha = 1/m\omega_c$. We find

$$\frac{1}{g_c}q_a^p + \frac{\imath}{g_c}\eta s_p - \sum_n \big(E_p - \varepsilon_n - q_a^p\big)^{-1} = 0, \tag{3.10}$$

where $g_c = g/2\pi\alpha$, $1/2\pi\alpha$ = degeneracy per Landau level. This equation has been obtained previously by Ando [16] using the self-consistent Born approximation (SCBA). The solution is of the form

$$q_a^p = e_0 + \imath s_p \rho_0, \tag{3.11}$$

where ρ_0 is proportional to the density of states in SCBA.

As the theory is invariant under $U(m, m)$ for $\eta = 0$, the most important fluctuations will come from those Q which take the form $Q \to T^{-1}Q_{sp}T = e_0 + \imath\rho_0 T^{-1}sT$ for $T \in U(m, m)$. For $T \in U(m) \times U(m)$ the saddle point remains unchanged and we expect these fluctuations to be unimportant. These statements can be formalized using the representation

$$Q(r) = T^{-1}(r)P(r)T(r), \tag{3.12}$$

where

$$P(r) = \delta_{pp'}P^{p}_{ab}(r) = P^{+}(r),$$

$$T(r) = T^{pp'}_{ab}(r) = T^{+}(r), \qquad T \in U(m, m). \tag{3.13}$$

The condition of hermitian T serves as to divide out the factor group $U(m) \times U(m)$. This split in fields of integration acquires a jacobian $I[P]$ for the longitudinal components P

$$I[P] = \prod_{a,b}\left(\lambda^{(1)}_a - \lambda^{(2)}_b\right)^2, \tag{3.14}$$

where λ^{p}_a are the eigenvalues of P^{p}_{ab}.

In what follows, the strategy [1, 2] will be to obtain an effective lagrangian in the massless fields T by formally integrating out the massive modes P

$$\exp \hat{L}[T] \equiv \int D[P] \exp L[P, T], \tag{3.15}$$

where

$$L[P, T] = \mathrm{Tr}\ln[I[P]] - \frac{1}{2g}\mathrm{Tr}\,P^2 - \imath\frac{\eta}{g}\mathrm{Tr}[TsT^{-1}]P - \mathrm{Tr}\ln\left[E - T\pi_\mu\pi^\mu T^{-1} - P\right]. \tag{3.16}$$

For this purpose we split L into two parts

$$L[P, T] = L_0[P] + \delta L[P, T], \tag{3.17}$$

with

$$L_0[P] = \mathrm{Tr}\ln[I[P]] - \frac{1}{2g}\mathrm{Tr}\,P^2 - \imath\frac{\eta}{g}\mathrm{Tr}\,sP - \mathrm{Tr}\ln\left[E - \pi_\mu\pi^\mu - P\right],$$

$$\delta L[P, T] = L[P, T] - L_0[P], \tag{3.18}$$

such that $\hat{L}[T]$ can be obtained by cumulant-expanding $\delta L[P, T]$ with respect to

$L_0[P]$. For $\eta = 0$, δL in turn is obtained from expanding the Tr ln with respect to the quantity

$$B[T] = T\pi_\mu \pi^\mu T^{-1} - \pi_\mu \pi^\mu, \tag{3.19}$$

which effectively proceeds as a gradient expansion in the field T. In order to study the critical properties of the system near $d = 2$ one needs therefore to consider the first few cumulants with respect to L_0 only.

Introducing the symbol

$$\partial_\mu \equiv \frac{-1}{\imath\sqrt{2m}} \overleftarrow{\nabla}_\mu, \qquad \partial^\mu \equiv \frac{+1}{\imath\sqrt{2m}} \overrightarrow{\nabla}_\mu, \tag{3.20}$$

then $B[T]$ can be written

$$B[T] = D_\mu D^\mu + D_\mu \pi^\mu + \pi_\mu D^\mu,$$

$$D^\mu = D_\mu = T\partial_\mu T^{-1}. \tag{3.21}$$

It is important to notice at this point that $\hat{L}$ as defined by (3.15) should be rigorously invariant under $T(r) \to U(r)T(r)$ with $U \in \mathrm{U}(m) \times \mathrm{U}(m)$. This symmetry, however, seems to be broken by the gradient expansion, namely under such a transformation D_μ goes over into

$$D^\mu \to U[D^\mu - U^{-1}\partial^\mu U]U^{-1}. \tag{3.22}$$

In order to resolve this question it is important, once more, that the correlations described by $L_0[P]$ are short-ranged. This will be checked in subsect. 3.2.

3.2 LONGITUDINAL MODES P

Expanding L_0 (3.18) around the saddle point (3.10) and keeping only the surviving terms when $m \to 0$ we find

$$L_0[P] = -\mathrm{Tr}\, P^p_{ab}(r) M^{pp'}_{ab,cd}(r - r') P^{p'}_{cd}(r') + \mathrm{O}(P^3), \tag{3.23}$$

with

$$M^{pp'}_{ab,cd}(r - r') = \delta_{ad}\delta_{bc}\delta_{pp'} M^p_0(r - r') + \delta_{ab}\delta_{cd}(1 - \delta_{pp'}) \frac{1}{2\rho_0^2} \delta_{rr'},$$

$$M^p_0(r - r') = \frac{1}{2g}\delta_{rr'} - \tfrac{1}{2} G^p_0(r, r') G^p_0(r', r),$$

$$G^p_0(r, r') = \langle r | \left(E - \pi_\mu \pi^\mu - q^p\right)^{-1} | r' \rangle$$

$$= \sum_{k,n} \frac{\phi_n(x + k\alpha)\phi_n^*(x' + k\alpha)}{E - \varepsilon_n + q^p} e^{\imath k(y - y')}. \tag{3.24}$$

In (3.24) we employed the fact that the combination $G_0^p(r', r)$ is translationally invariant; the second term in M originates from the jacobian $I[P]$. The matrix inverse of M is easily obtained in the limit $m = 0$. In momentum representation one finds

$$\left[M^{-1}(q)\right]_{ab,\,cd}^{pp'} = \delta_{ad}\delta_{bc}\delta_{pp'}\Lambda^p(q) + \delta_{ab}\delta_{cd}(1 - \delta_{pp'})\chi(q), \tag{3.25}$$

where

$$\Lambda^p(q) = \left[M_0^p(q)\right]^{-1},$$

$$\chi(q) = \frac{1}{2\rho_0^2}\left[M_0^{(1)}(q)\,M_0^{(2)}(q)\right]^{-1}. \tag{3.26}$$

In order to establish the validity of a loop expansion in P it follows that we must show that $\mathrm{Re}\, M_0^p(q) > 0$ for all q. This can be done by noting that $M_0^p(q)$ can be written as

$$M_0^p(q) = \frac{1}{2g} - \frac{1}{4\pi\alpha}\sum_{n,n'}|F_{n,n'}(q)|^2 D_{n,n'}^p,$$

$$D_{n,n'}^p = \left(E - \varepsilon_n - q^p\right)^{-1}\left(E - \varepsilon_{n'} - q^p\right)^{-1},$$

$$F_{n,n'}(q) = \int_{-\infty}^{\infty} \mathrm{d}x\, \phi_{n'}^*(x)\phi_n(x)\mathrm{e}^{iqx}. \tag{3.27}$$

Hence

$$2\,\mathrm{Re}\, M_0^p(q) = \frac{1}{g} - \frac{1}{2\pi\alpha}\sum_n \left(E - \varepsilon_n - q^{(1)}\right)^{-1}\left(E - \varepsilon_n - q^{(2)}\right)^{-1}$$

$$+ \frac{1}{4\pi\alpha}\sum_{n,n'}|F_{n,n'}(q)|^2|\tilde{D}_{n,n'}|^2, \tag{3.28}$$

with

$$\tilde{D}_{n,n'} = \left(E - \varepsilon_n - q^{(1)}\right)^{-1} - \left(E - \varepsilon_{n'} - q^{(2)}\right)^{-1}. \tag{3.29}$$

From the saddle-point equation (3.10) it follows that the first two terms on the right-hand side of (3.28) cancel. It follows that $\mathrm{Re}\, M_0^p(q) > 0$ for all energies at which the density of states does not vanish.

The existence of a loop expansion in P allows us to formally carry on the evaluation of $\hat{L}[T]$.

3 3 TRANSVERSE MODES T

The expansion of the Tr ln in $\delta L[P, T]$ in (3.18) proceeds in terms of propagators of the form

$$G^p_{ab}(r, r') = \langle r | \Big[\big(E - \pi_\mu \pi^\mu - P(r) \big)^{-1} \Big]^p_{ab} | r' \rangle , \tag{3.30}$$

which are to be weighted with respect to $L_0[P]$ (3.18).

Expanding up to second order in $B[T]$ (3.18) and to first order in η we find

$$\begin{aligned}
\hat{L}[T] = & \frac{-i\eta}{g} \mathrm{Tr}[TsT^{-1} - s]^{pp}_{ab} \langle P^p_{ba}(r) \rangle \\
& + \mathrm{Tr}\big(D_\mu D^\mu \big)^{pp}_{ab} \langle G^p_{ba}(r, r) \rangle \\
& + \mathrm{Tr}\big[D_\mu(r) \big]^{pp}_{ab} \langle \big(\pi_\mu + \pi^\mu \big) G^p_{ba}(r, r) \rangle \\
& + 2\,\mathrm{Tr}\big[D_\mu(r) \big]^{pp'}_{cb} \big[D_\nu(r') \big]^{p'p}_{ad} \langle G^p_{dc}(r', r) \pi^\mu G^{p'}_{ba}(r, r') \pi^\nu \rangle \\
& + 2\,\mathrm{Tr}\big[D_\mu(r) \big]^{pp}_{ab} \big[D_\nu(r') \big]^{p'p'}_{cd} \langle \big[\pi^\mu G^p_{ba}(r, r) \big] \big[\pi^\nu G^{p'}_{dc}(r', r') \big] \rangle_c ,
\end{aligned} \tag{3.31}$$

where the statistical average $\langle \ldots \rangle$ is with respect to L_0 and with D_μ given by (3.21).

3.3.1. Gradient expansion. All averages appearing in (3.31) are translationally invariant and as discussed previously, short-ranged. We may, therefore, carry the expansion one step further by gradient expanding the propagator averages, e.g.

$$\begin{aligned}
\langle G^p_{ad}(r', r) \pi_\mu G^{p'}_{bc}(r, r') \pi_\nu \rangle &= K^{pp'}_{ad,\,bc;\,\mu\nu}(r - r') \\
&= K^{(0)} \delta_{rr'} + K^{(2)} \delta_{rr'} \nabla_i \nabla_j + \cdots ,
\end{aligned} \tag{3.32}$$

where the $K^{(i)}$ are matrices. In this way we can collect the terms with at most two gradients in the field T.

A further simplification occurs due to the fact that $L_0[P]$ is invariant under global $\mathrm{U}(m) \times \mathrm{U}(m)$ which implies that the expectations vanish unless the replica indices coincide pairwise as

$$\begin{aligned}
& \langle P^p_{ab}(r) \rangle = \delta_{ab} g \tilde{g}^p , \qquad \langle G^p_{ab}(r, r) \rangle = \delta_{ab} \tilde{g}^p , \\
& \langle \big(\pi_\mu + \pi^\mu \big) G^p_{ab}(r, r) \rangle = \delta_{ab} \tilde{g}^p_\mu , \\
& \sum_r \langle G^p_{dc}(r', r) \pi^\mu G^{p'}_{ba}(r, r') \pi^\nu \rangle = \delta_{cd} \delta_{ba} K^{pp'}_{0,\mu\nu} + \delta_{cb} \delta_{ad} \delta_{pp'} K^p_{1,\mu\nu} , \\
& \sum_r \langle \big[\pi^\mu G^p_{ba}(r, r) \big] \big[\pi^\nu G^{p'}_{dc}(r', r') \big] \rangle_c = \delta_{ab} \delta_{dc} C^{pp'}_{0,\mu\nu} + \delta_{bc} \delta_{ad} \delta_{pp'} C^p_{1,\mu\nu} .
\end{aligned} \tag{3.33}$$

Collecting the results we may write for $\hat{L}$

$$\hat{L}[T] = \mathrm{Tr} + \tilde{g}_{\mu}^{p}\left[D_{\mu}\right]_{aa}^{pp} + \hat{k}_{\mu\nu}^{pp'}\left[D_{\mu}\right]_{ab}^{pp'}\left[D_{\nu}\right]_{ba}^{p'p} + \tilde{k}_{\mu\nu}^{pp'}\left[D_{\mu}\right]_{aa}^{pp}\left[D_{\nu}\right]_{bb}^{p'p'}$$

$$+ \mathrm{O}\left(D_{\mu}^{3}\right) - i\tilde{g}^{p}\eta\left[TsT^{-1} - s\right]_{aa}^{pp}, \tag{3.34}$$

where

$$\hat{k}_{\mu\nu}^{pp'} = \delta_{\mu\nu}\left(\frac{\tilde{g}^{p} + \tilde{g}^{p'}}{2}\right) + 2K_{0,\mu\nu}^{pp'} + 2C_{1,\mu\nu}^{p}\delta_{pp'}, \tag{3.35}$$

$$\tilde{k}_{\mu\nu}^{pp'} = 2C_{0,\mu\nu}^{pp'} + 2K_{1,\mu\nu}^{p}\delta_{pp'}. \tag{3.36}$$

3.3.2. Analysis via Ward identities and the effect of boundaries. A further reduction of $\hat{L}$ occurs by noting the invariance of the underlying theory defined by $L_0[P]$ under local $\mathrm{U}(m)\times\mathrm{U}(m)$ rotations, i.e. invariance under (3.22) implies the vanishing of the following quantities

$$\hat{k}_{\mu\nu}^{pp} = 0, \qquad \tilde{k}_{\mu\nu}^{pp'} = 0. \tag{3.37}$$

A more careful formulation of this statement is given in the appendix. The term linear in D_{μ} in (3.34) however needs special consideration. In the appendix (A.7) we conclude that the quantity $\tilde{g}_{\mu}^{p}$ vanishes in the interior of the system. However, this conclusion might change if one imposes boundaries on the system (as already indicated in subsect. 2.3). In other words, the term linear in D_{μ} in (3.34) might contribute as a surface term and hence, formally contribute as a second-order term in the gradient expansion.

In the following we will show how the occurrence of this possible boundary effect emerges from the properties of the interior of the sample. We rewrite the expectation of the current as

$$\tilde{g}_{\mu}^{p} = \int_{-\infty}^{E} \mathrm{d}E^{p}\,\frac{\partial}{\partial E^{p}}\left\langle\left(\pi_{\mu} + \pi^{\mu}\right)G_{bb}^{p}(r,r)\right\rangle$$

$$= \int_{-\infty}^{E} \mathrm{d}E^{p}\sum_{c,r'}\left\{\left\langle\left(\pi_{\mu} + \pi^{\mu}\right)G_{bc}^{p}(r,r')G_{cb}^{p}(r',r)\right\rangle\right.$$

$$\left. - \left\langle\left[\left(\pi_{\mu} + \pi^{\mu}\right)G_{bb}^{p}(r,r)\right]\left[G_{cc}^{p}(r',r')\right]\right\rangle_{\mathrm{cum}}\right\}, \tag{3.38}$$

where the variations in energy E^{p} are being performed for $p = 1, 2$ independently. We proceed as before by evaluating the two-particle Green functions in (3.38) for an infinite system. These quantities are then short-ranged and translational invariant.

We make use of this notion and Taylor-expand $D_\mu(r)$ about the point r', that is

$$\mathrm{Tr}\,\tilde{g}^p_\mu(r)\big[D_\mu\big]^{pp}_{aa}(r)$$

$$= \int_{-\infty}^{E} \mathrm{d}E^p \sum_{a,p} \sum_{r'} \sum_{r} \Big\{ \big[D_\mu(r')\big]^{pp}_{aa} + (r-r')_\nu \big[\nabla_\nu D_\mu(r')\big]^{pp}_{aa} + \cdots \Big\} \times \Lambda^p_\mu(r-r'),$$

$$\Lambda^p_\mu(r-r')$$

$$= \sum_{c} \langle (\pi_\mu + \pi^\mu) G^p_{bc}(r,r') G^p_{cb}(r',r)\rangle - \langle \big[(\pi_\mu + \pi^\mu) G^p_{bb}(r,r)\big]\big[G^p_{cc}(r',r')\big]\rangle_{\mathrm{cum}}.$$

(3.39)

We now make use of the identities derived in the appendix. Namely from (A.11), in the appendix, we have

$$\sum_{r} \langle (\pi_\mu + \pi^\mu) G^p_{bc}(r,r') G^p_{cb}(r',r)\rangle = 0.$$

Furthermore, the general result (A.8) also implies

$$\sum_{r} \langle \big[(\pi_\mu + \pi^\mu) G^p_{bb}(r,r)\big]\big[G^p_{cc}(r',r')\big]\rangle_{\mathrm{cum}} = 0. \tag{3.40}$$

Hence the lowest-order term in the Taylor series expansion vanishes. The problem that remains is to evaluate the coefficient of the second-order term $\nabla_\nu D_\mu$ in (3.39), that is

$$\mathrm{Tr}\,\tilde{g}^p_\mu(r)\big[D_\mu(r)\big]^{pp}_{aa} = \sum_{p,a} \sum_{r'} \big[\partial^\nu D_\mu(r')\big]^{pp}_{aa} \Big\{ \imath\sqrt{2m} \int_{-\infty}^{E} \mathrm{d}E^p \sum_{r} r_\nu \Lambda^p_\mu(r-r') \Big\}.$$

(3.41)

Notice that identity (A.8) in the appendix further restricts the expression. Namely the only contribution comes from the combination $\varepsilon_{\mu\nu} r_\mu \Lambda_\nu$ inside the expectation brackets. For all other combinations the coefficient multiplying the Λ_μ can be expressed as a total derivative and therefore vanish after integrating over r. This, again is an exact statement. We are left with

$$\mathrm{Tr}\,\tilde{g}^p_\mu(r)\big[D_\mu(r)\big]^{pp}_{aa} = \sum_{p} \xi^p \sum_{r',a} \big[\varepsilon_{\mu\nu}\partial^\nu D_\mu(r')\big]^{pp}_{aa} + \mathrm{O}\big(\partial^2 D_\mu\big), \tag{3.42}$$

with

$$\xi^p = -\tfrac{1}{2}\imath\sqrt{2m}\int_{\infty}^{L} \mathrm{d}E^p \sum_{r,c}\Big\{\langle \varepsilon_{\mu\nu} r_\nu \pi_\mu G^p_{bc}(r,r') G^p_{cb}(r',r)\rangle$$

$$-\langle\big[\varepsilon_{\mu\nu} r_\nu \pi_\mu G^p_{bb}(r,r)\big]\big[G^p_{cc}(r',r')\big]\rangle_{\mathrm{cum}}\Big\}$$

$$= -\imath mc/e \times \int_{-\infty}^{L} \mathrm{d}E^p \frac{\partial}{\partial B^p}\langle G^p_{dd}(r',r')\rangle, \tag{3.43}$$

where B^p indicates that the variation in magnetic field is to be done in the $p = 1, 2$ subspaces separately. The last step can most easily be understood in the symmetric gauge.

In the limit of zero replicas the quantity ξ^p are the exact matrix elements that enter the expression for the Hall conductance (i.e. $\sigma^{\mathrm{II}}_{xy}$) as derived by Streda [5].

The expression above enters the lagrangian despite the fact that it can be written as a total divergence and therefore naively does not contribute. However, all the physics of the problem is contained in this term [9, 10]. The statistical quantity

$$\Phi = \mathrm{Tr}\,\varepsilon_{\mu\nu}\partial^\mu D_\nu s, \tag{3.44}$$

multiplying the conductance

$$\sigma^{\mathrm{II}}_{xy} \equiv \tfrac{1}{2}\big(\xi^1 - \xi^2\big), \tag{3.45}$$

can be interpreted as a fluctuation in magnetic flux away from an effective medium theory of impurity scattering. It is very close to the spirit of Laughlin's work [6].

The derivative acting on D_μ above can be written as

$$\mathrm{Tr}\,2\tilde{g}^p_\mu \big[D_\mu\big]^{pp}_{aa} = \mathrm{Tr}\big[\sigma^{\mathrm{II}}_{xy}\varepsilon_{\mu\nu}\varepsilon_{pp'}\big]\big[D_\mu\big]^{pp'}_{ab}\big[D_\nu\big]^{p'p}_{ba} + \mathrm{O}\big(D^3_\mu\big). \tag{3.46}$$

Using this result, together with eq. (3.37), we can write for eq. (3.34)

$$\hat{L}[T] = \mathrm{Tr}\,\rho^{pp'}_{\mu\nu}\big[D_\mu\big]^{pp'}_{ab}\big[D_\nu\big]^{p'p}_{ba} - \imath\tilde{g}^p\eta\big[TsT^{\,1} - s\big]^{pp}_{aa}, \tag{3.47}$$

with

$$\rho^{pp'}_{\mu\nu} = \hat{k}^{pp'}_{\mu\nu}(1 - \delta^{pp'}) + \sigma^{\mathrm{II}}_{xy}\varepsilon_{\mu\nu}\varepsilon_{pp'}. \tag{3.48}$$

This expression finally is rigorously symmetric under $\mathrm{U}(m) \times \mathrm{U}(m)$.

3.3.3. Further evaluation. Notice that when the Green functions are approximated by their saddle-point values, ρ reduces to the magneto conductivity as

calculated in SCBA. More generally, the $\hat{k}$ can be identified indeed with a classical conductivity tensor. In order to see this, recall that [see eq. (3.33)]

$$K^{12}_{0,\mu\nu} = \sum_{r'} \langle G^{(1)}_{aa}(r,r')\pi_\mu(r')G^{(2)}_{bb}(r',r)\pi_\nu(r)\rangle . \tag{3.49}$$

The complex conjugate of (3.48) can be evaluated to give

$$K^{12*}_{0,xy} = -K^{21}_{0,xy}. \tag{3.50}$$

It is, furthermore, easily established that K_0 is real symmetric

$$K_0 = K_0^{\mathrm{T}}. \tag{3.51}$$

Let us first concentrate on the terms in $\hat{L}$ diagonal in x, y. From eqs. (3.51) and (3.37) it follows

$$\begin{aligned}\sigma_{0,\mu\mu} &= \hat{k}^{12}_{\mu\mu} = \hat{k}^{21}_{\mu\mu} \\ &= K^{12}_{0,\mu\mu} + K^{21}_{0,\mu\mu} - \left(K^{11}_{0,\mu\mu} + C^{1}_{1,\mu\mu}\right) - \left(K^{22}_{0,\mu\mu} + C^{2}_{1,\mu\mu}\right),\end{aligned} \tag{3.52}$$

which has the form of a current-current correlation function.

The part containing $\sigma_{0,\mu\mu}$ can be casted in the form

$$\hat{L}_{\mu\mu} \equiv \sigma_{0,\mu\mu}\mathrm{Tr}\left[D^{12}_{\mu}D^{21}_{\mu} + D^{21}_{\mu}D^{21}_{\mu}\right] = -\tfrac{1}{2}\sigma_{0,\mu\mu}\mathrm{Tr}\,\partial^\mu\tilde{Q}\partial^\mu\tilde{Q}, \tag{3.53}$$

where

$$\tilde{Q} = T^{-1}sT. \tag{3.54}$$

Eq. (3.53) constitutes, in more conventional notation, the U(m, m) non-linear σ-model.

Next we consider the off-diagonal parts in $\hat{L}$. From (3.37), (3.39) and (3.40) it follows

$$\sigma^{\mathrm{I}}_{0,xy} = \hat{k}^{12}_{xy} = -\hat{k}^{12}_{yx} = K^{12}_{0,xy} - K^{12}_{0,yx}, \tag{3.55}$$

which can be interpreted as the classical contribution to the Hall conductivity and which in its simplest form (SCBA or saddle-point approximation) reduces to the Drude-Zener formula [5]

$$\sigma_{0,xy} = \omega_c\tau\sigma_{0,\mu\mu}, \tag{3.56}$$

where the relaxation time τ is identified with ρ_0^{-1} (3.11).

The part in $\hat{L}$ containing $\sigma^{\mathrm{I}}_{0,xy}$ and $\sigma^{\mathrm{II}}_{xy}$ can be written

$$\hat{L}_{xy} \equiv 2\sigma_{0,xy}\mathrm{Tr}\left[D^{12}_{x}D^{21}_{y}\right] - \left[D^{21}_{x}D^{12}_{y}\right] = -\tfrac{1}{4}\sigma_{0,xy}\mathrm{Tr}\,\tilde{Q}\left[\partial^{x}\tilde{Q}, \partial^{y}\tilde{Q}\right], \quad (3.57)$$

with

$$\sigma_{0,xy} = \sigma^{\mathrm{I}}_{0,xy} + \sigma^{\mathrm{II}}_{xy}. \quad (3.58)$$

Written as a total divergence we have in terms of T

$$\hat{L}_{xy} = \sigma_{0,xy}\mathrm{Tr}\,\partial^{\mu}\left[\varepsilon_{\mu\nu}T\partial^{\nu}T^{-1}s\right]. \quad (3.59)$$

We complete the derivation by including the symmetry-breaking term

$$\hat{L}_{\eta} = -\eta\pi\rho(E)\mathrm{Tr}\,s(\tilde{Q}-s). \quad (3.60)$$

Notice that this term contains the exact expression for the density of states which is a necessary feature [2]. The results of this section can be summarized as

$$\hat{L}[T] = -\tfrac{1}{2}\sigma_{0,\mu\mu}\mathrm{Tr}\,\partial^{\mu}\tilde{Q}\partial^{\mu}\tilde{Q} - \tfrac{1}{4}\sigma_{0,xy}\mathrm{Tr}\,\tilde{Q}\left[\partial^{x}\tilde{Q}, \partial^{y}\tilde{Q}\right] - \eta\pi\rho(E)\mathrm{Tr}\,s(\tilde{Q}-s), \quad (3.61)$$

where the $\tilde{Q}$ belongs to the non-compact group $\mathrm{U}(m,m)/\mathrm{U}(m)\times\mathrm{U}(m)$. Finally, after one takes into account the proper dimension of the operators, the σ's entering eq. (3.61) are dimensionless numbers, measuring the conductivity in units of $2e^2/h$.

3.4 COMPACT SYMMETRIES

In terms of fermion fields the analysis proceeds almost identically but there are subtle differences.

We start from eq. (3.1), with a different meaning for the $\bar{\Psi}, \Psi$ (see sect. 2). The decoupling of the four-point term can now be performed introducing a hermitian Q which preserves the $\mathrm{U}(2m)$ symmetry. Taking into account the anti-commutativity, eq. (3.6) now becomes

$$L = -\frac{1}{2g}\mathrm{Tr}\,Q^2 + \mathrm{Tr}\ln\left[E_p - \pi_{\mu}\pi^{\mu} - \imath Q\right]. \quad (3.62)$$

It is easily checked that this lagrangian gives rise to an identical saddle point except for a factor $\imath$. One has to make sure that the saddle point can be reached by a continuous deformation of the path of integration for the diagonal components of Q such that one does not hit the singularities of the Tr ln. However, one can always

define the path for the diagonal components such that these singularities are avoided and at the same time eq. (3.62) still holds. The reason is that these components are redundant in the sense that they are associated with the terms in the four-point coupling which have identical indices and which vanish because of the Grassmann algebra.

One can now formally proceed by splitting the Q into massive components P and critical ones as in (3.12), the T now belonging to U($2m$). Under the following transformation

$$P \to iP, \qquad L \to -L, \qquad \mathrm{U}(m,m) \to \mathrm{U}(2m), \tag{3.63}$$

the previous analysis is translated in compact symmetries.

As a final remark, we emphasize here that the compact symmetries in this case are dictated by convergence requirements in the same way, although not so transparent from the outset, as non-compact symmetries emerge from the boson representation. That is, naively one could have introduced also pseudo-hermitian Q (3.4) in the fermion representation. This however, would lead to serious convergence problems, the transverse fluctuations around the saddle point then have the wrong sign or, following up the previous analysis, one ends up with the wrong sign of the coupling and symmetry-breaking term in the final lagrangian.

4. Discussion

The term with $\sigma_{0,xy}$ in $\hat{L}$, for which a constant magnetic background field is directly responsible, forms a new feature in the theory of electronic disorder.

In the context of the compact symmetries, the quantity $\frac{1}{8}\mathrm{Tr}\,\tilde{Q}[\partial_x\tilde{Q}, \partial_y\tilde{Q}]$ in its explicit form has been noted before as the topological invariant labeling the different topological sectors of the $\mathrm{U}(m+n)/\mathrm{U}(m)\times\mathrm{U}(n)$ theory by integer quantum numbers [18,19]. It is argued [9,10] that the presence of such a term in the lagrangian leads to drastic changes in the β function as obtained from the perturbative renormalization group approach [17] and, in fact, is capable of changing the long-distance behaviour of the theory; for the disordered electronic system this has the meaning of delocalization of the electronic state. These arguments emphasize a direct analogy with four-dimensional Yang-Mills theory with $\sigma_{0,xy}$ replacing the instanton- or vacuum-angle θ.

Contrary to the situation in particle physics, the instanton angle θ or $\sigma_{0,xy}$ appears for the 2d electronic system as an experimentally adjustable parameter changing with the location of the Fermi level, concentration of impurities and magnetic field strength.

The effect has been studied in the dilute instanton gas approximation and will be published elsewhere [10]. Here we only stress the result that the effects of instantons persist also in the case of interest, namely the limit of zero number of field

components. More general arguments, based on 't Hooft's duality idea, lead to the existence of extended states for special values of $\sigma_{0,xy}$. These special values occur at the center of each Landau level namely where $\sigma_{0,xy}$ passes through half-integer values (in units of e^2/h) [10].

Away from the band center, the electronic states are exponentially localized. The occurrence of a Hall conductance in this case is due to the existence of current-carrying states near the boundary of the system. The quantization of the Hall conductance (in units e^2/h) can then be shown to occur on topological grounds [10]. Therefore, the picture that emerges is theoretically complete and consistent with the experiments on the quantum Hall effect.

Finally, as far as the non-compact symmetries are concerned, the problem is unresolved. The perturbative β functions for compact and non-compact σ-models are simply related by a sign change in the coupling [15]. For the unitary non-linear σ-model they lead to identical results in the analytic continuation of zero components [17], showing the important feature of asymptotic freedom. It is because of this latter observation that one can justify the analysis mentioned above in terms of compact symmetries. The non-compact model, on the other hand, is not asymptotically free for $\sigma_{0,xy} = 0$ and for all $n + m > 0$ and a translation of the topology-based arguments is not obvious: there are no finite-action solutions to the classical equations of motion in this case, i.e. there are no instantons [20] and it is not clear whether a smooth extrapolation of some kind does exist. More work is needed in order to understand this aspect of the problem.

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Appendix

Consider the lagrangian

$$L = \mathrm{Tr}\,\bar{\Psi}_\alpha^{(p)}(r)\left[\left(E - \pi_\mu\pi^\mu\right)\delta_{\alpha\beta}^{pp'} - P_{\alpha\beta}^{p}(r) + h_{\alpha\beta}^{pp'}(rr')\right]\Psi_\beta^{p'}(r'). \tag{A.1}$$

We will study the consequences of invariance under $\mathrm{U}(m)\times\mathrm{U}(m)$. We introduce a symbol $\tilde{h} = \tilde{h}^+$

$$\tilde{h}_{\alpha\beta}^{pp'}(rr') = \delta_{rr'}\delta^{pp'}\delta_{\alpha\beta}\pi_\mu(r)\pi^\mu(r') + h_{\alpha\beta}^{pp'}(rr'), \tag{A.2}$$

using the convention of eq. (3.20) for the operators $\pi_\mu\pi^\mu$. Then certainly ($F = \ln Z = \ln \int e^L$)

$$F[\tilde{h}] = F\left[U^{-1}(r)\tilde{h}(rr')U(r')\right], \tag{A.3}$$

where $U(r) \equiv \delta^{pp'} U^p_{\alpha\beta}(r)$. For an infinitesimal unitary transformation

$$U = 1 + \eta\varepsilon, \qquad \varepsilon^+ = -\varepsilon, \qquad \eta = \text{infinitesimal},$$

$$F[\tilde{h}] = F[\tilde{h} + \eta[\tilde{h}, \varepsilon]], \tag{A.4}$$

we get

$$0 \equiv \frac{\partial}{\partial\eta} F = \frac{\partial F}{\partial \tilde{h}} \frac{\partial \tilde{h}}{\partial \eta} = \frac{\partial F}{\partial \tilde{h}}[\tilde{h}, \varepsilon]$$

$$= \mathrm{Tr} \frac{\partial F}{\partial \tilde{h}^{pp'}_{\alpha\beta}(rr')} \left[\tilde{h}^{pp'}_{\alpha\gamma}(rr') \varepsilon^{p'}_{\gamma\beta}(r') - \varepsilon^{p}_{\alpha\gamma}(r) \tilde{h}^{pp'}_{\gamma\beta}(rr') \right]. \tag{A.5}$$

(i) This expression, evaluated at $h = 0$ gives

$$\sum_{rr'} \left(\partial^\mu \varepsilon^p_{\alpha\beta}(r) \right) \delta_{rr'} \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \frac{\partial F}{\partial h^{pp}_{\alpha\beta}(rr')} \right] = 0, \tag{A.6}$$

which should hold for any $\varepsilon = -\varepsilon^+$. Choose $\varepsilon(r) = i\delta_{\alpha\beta}\delta_{\alpha a} r_\mu$ then it follows

$$\sum_{rr'} \delta_{rr'} \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \frac{\partial F}{\partial h^{pp}_{aa}(rr')} \right] \equiv \langle \bar{\Psi}^{(p)}_a(r) \left(\pi_\mu + \pi^\mu \right) \Psi^{(p)}_a(r) \rangle = 0. \tag{A.7}$$

That is, the current has zero expectation value or $\tilde{g}^p_\mu = 0$ in the interior of the system.

(ii) Differentiating eq. (A.5) with respect to $\tilde{h}$ leads to

$$0 = \sum_{p,\, \alpha\beta,\, rr'} \delta_{rr'} \partial^\mu \varepsilon^p_{\alpha\beta}(r) \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \frac{\partial^2 F}{\partial h^{pp}_{\alpha\beta}(rr') \, \partial h^{qq}_{ab}(\rho\rho')} \right]$$

$$+ \sum_\beta \left(\frac{\partial F}{\partial h^{qq}_{a\beta}(\rho\rho')} \varepsilon^q_{b\beta}(\rho') - \frac{\partial F}{\partial h^{qq}_{\beta b}(\rho\rho')} \varepsilon^q_{\beta a}(\rho) \right), \tag{A.8}$$

valid for any function $\varepsilon(\rho) = -\varepsilon^+(\rho)$.

Specifically, taking $\varepsilon^p_{\alpha\beta}(r) = \delta_{\alpha,a}\delta_{\beta,b}\delta^{p,q}\varepsilon r_\mu - \delta_{\alpha,b}\delta_{\beta,a}\delta^{p,q}\varepsilon^* r_\mu$ with ε arbitrary complex, we get

$$\sum_{rr'} \delta_{rr'} \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \left(\varepsilon \frac{\partial^2 F}{\partial h^{qq}_{ab}(rr') \partial h^{qq}_{ab}(\rho\rho')} - \varepsilon^* \frac{\partial^2 F}{\partial h^{qq}_{ba}(rr') \partial h^{qq}_{ab}(\rho\rho')} \right) \right]$$

$$- \frac{\partial F}{\partial h^{qq}_{aa}(\rho\rho')} \varepsilon^* \rho_\mu + \frac{\partial F}{\partial h^{qq}_{bb}(\rho\rho')} \varepsilon^* \rho_\mu = 0, \qquad [a \neq b], \tag{A.9}$$

which tells us ($a \neq b$)

$$\sum_{rr'} \delta_{rr'} \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \frac{\partial^2 F}{\partial h_{ab}^{qq}(rr') \partial h_{ab}^{qq}(\rho\rho')} \right] = 0, \tag{A.10}$$

$$\sum_{rr'} \delta_{rr'} \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \frac{\partial^2 F}{\partial h_{ba}^{qq}(rr')\, \partial h_{ab}^{qq}(\rho\rho')} \right] + \left(\rho'_\mu - \rho_\mu \right) \frac{\partial F}{\partial h_{aa}^{qq}(\rho\rho')} = 0. \tag{A.11}$$

Operating the LHS and the RHS with $\delta_{\rho\rho'}(\pi_\mu(\rho) + \pi^\mu(\rho'))$ finally gives [$\mu \neq \nu$]

$$\sum_{rr'} \delta_{rr'} \delta_{\rho\rho'} \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \left(\pi_\nu(\rho) + \pi^\nu(\rho') \right) \frac{\partial^2 F}{\partial h_{ba}^{qq}(rr')\, \partial h_{ab}^{qq}(\rho\rho')} \right] = 0, \tag{A.12}$$

which tells us directly that [see (3.35)] $\hat{k}_{xy}^{pp} = \hat{k}_{yx}^{pp} = 0$

$$\sum_{rr'} \delta_{rr'} \delta_{\rho\rho'} \left[\left(\pi_\mu(r) + \pi^\mu(r') \right) \left(\pi_\mu(\rho) + \pi^\mu(\rho') \right) \frac{\partial^2 F}{\partial h_{aa}^{qq}(rr') \partial h_{ab}^{qq}(\rho\rho')} \right] + \frac{\partial F}{\partial h_{aa}^{qq}(\rho\rho)} = 0, \tag{A.13}$$

from which follows directly [see (3.35)] $\hat{k}_{\mu\mu}^{pp} = 0$.

Finally by taking the ε in (A.8) diagonal in the replica indices it follows immediately that the quantity $\tilde{k}$ in (3.36) vanishes.

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